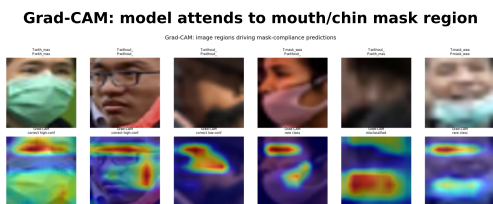


Graphical Abstract

An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery

Shivaram Balla



Three-Class Face-Mask Compliance

Pipeline

1. Kaggle FMD: 853 images, PASCAL-VOC
2. Crop faces -> 4,072 crops, 3 classes
3. Softened class weights (5.1/1.0/2.1)
4. Custom CNN (259k params, 64x64)
5. Grad-CAM + SHAP + robustness

Key results

- Test accuracy: 92.5%
- with_mask F1 0.96, without_mask F1 0.88
- Rare incorrect-mask class hard (F1 0.32)
- Explanations focus on lower face
- Robust to blur; sensitive to heavy noise

Highlights

An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery

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- Three-class mask-compliance classification separates the high-risk incorrectly-worn case from mask and no-mask.
- A from-scratch custom CNN reaches 92.46% test accuracy on 4,072 cropped faces.
- Softened class weighting prevents minority-class collapse caused by severe imbalance.
- Grad-CAM and SHAP confirm the model attends to the mouth and chin mask region.
- Robustness testing reveals graceful blur tolerance but strong sensitivity to additive noise.

An Explainable and Robustness-Aware Convolutional Neural Network Framework for Three-Class Face-Mask-Compliance Classification from Public Surveillance Imagery

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Abstract

Face mask wear was identified as a critical, non-pharmaceutical remedy in the field of respiratory disease and computer vision-aware monitoring of mask compliance in public settings is an increasingly applicable problem.

But most published face-mask studies treat the problem as a binary mask versus no-mask classification while relatively few actually systematically assess the more difficult three-class setting that recognises a mask not worn correctly, which is in fact precisely the condition that undercuts the protective purpose of masking.

This study tackles a 3 class problem, namely, identifying masked individuals who are correctly worn (for example, the non-mask, cropped persons), incorrectly worn (one that has not been properly worn), and absent (those who have not been fitted with the mask as intended); it combines predictive assessment with explainability and robustness analysis rather than just reporting the accuracy data.

A custom convolutional neural network is trained with 4,072 face crops generated from the 853-image PASCAL-VOC-annotated Kaggle Face Mask Detection data via softened class weighting to alleviate extreme class imbalance.

With a matched stratified test set, the model returns a total accuracy of 92.46%, one weighted F1-score of 0.93, two well-represented classes ROC-AUC of 0.982 and 0.993 per-class. Grad-CAM and SHAP analyses confirm that predictions depend on the mouth and chin region vs background artefacts and robustness testing shows gentle degradation at blurring and occlu-

sion but high sensitivity to heavy additive noise.

The key contribution is the reproducible, explainable, and robustness aware baseline, which highlights that the exceptionality of the rare incorrectly-worn mask class is the greatest open challenge for deployment.

Keywords: Face mask detection, Convolutional neural networks, Transfer learning, Explainable AI, Image classification, Class imbalance

1. Introduction

1.1. Application Background and Practical Importance

Face masks were among the most popular non-pharmaceutical interventions for the recent respiratory-disease outbreaks, and public health authorities continued to rely on wearing them to reduce airborne transmission in crowded indoor spaces. In practice, the masks' effectiveness does not just depend on whether the mask is on, but whether it is worn correctly on both the nose and the mouth, because a mask worn incorrectly gives a much less protective effect and at the same time gives the appearance of safety. In high-throughput environments, e.g., airports, hospitals, public transport hubs, and large retail premises where compliance is monitored closely in visual form, manual visual checking for compliance of a mask is labor-intensive and inconsistent between human observers, which cannot be achieved at the scale needed for continuous monitoring. The people and organisations most impacted by this issue are safety-oriented facility operators in charge of ensuring that workers are safe and sound, frontline staff who are responsible for enforcing policies and the wider society as a whole where collective risk increases when compliance is poor. Delayed or erroneous evaluation of mask compliance can have a tangible impact, as not noticing non-compliance would lead to localized transmission clusters which may be expensive to address. Traditional manual methodologies also put enforcement officials face-to-face and repeatedly with people that may be non-compliant which exacerbates their threat. Automated image analysis presents an appealing alternative — the system can run continuously, make consistent decisions based on agreed criteria, and only call out the cases that need checking by humans. A good automated system can thus serve as a decision-support and prioritisation tool and not as a fully autonomous enforcement mechanism, the one that has been the frame adopted throughout the study.

29 1.2. Machine Learning and Computer Vision Context

30 Image-based machine learning has grown so quickly over the last 10 years
31 that convolutional neural networks have become the general paradigm for vi-
32 sual recognition problems. The key benefit of a convolutional neural network
33 is that it learns hierarchical visual features straight from pixels; this elimi-
34 nates the use of hand-crafted feature extraction methods commonly found in
35 previous computer-vision pipelines. Early convolutional layers usually model
36 low-level features like edges, colours, and textures, while deeper layers com-
37 bine elements of all of these structures into higher level patterns that can be
38 related to object parts and whole objects. In the face-mask task, this hierar-
39 chical ability is particularly useful, seeing as the difference between wearing
40 good masks, badly, or exposing faces is encoded in local texture, colour, and
41 the spatial organisation of the nose, mouth. In addition, convolutional net-
42 works are spatially equivariant with only minor changes of the position of the
43 respective facial region in the cropped image. However, being data-hungry,
44 the same flexibility which gives deep networks power leads them to overfit;
45 training a deep model from random initialisation typically necessitates lots
46 of labelled datasets in order to prevent overfitting. This limitation becomes
47 particularly evident for the face-mask domain, where publicly available an-
48 notated datasets are small and fairly imbalanced across the three compliance
49 classes. Accordingly, the methodological problems tackled by this study con-
50 cern the ability to achieve consistent convolutional capabilities under genuine
51 limitations of small and imbalanced datasets.

52 1.3. Transfer Learning for the Selected Problem

53 When a small number of images are labelled, it is hard to train a CNN
54 from random initialisation as the model has many parameters and even less
55 information from which to estimate them. Transfer learning solves this prob-
56 lem by reusing a model which trained on a very large general image set,
57 e.g., ImageNet dataset. The pre-trained backbone is capable of producing
58 general-purpose visual features - edge, texture and shape detectors - that are
59 useful across a large number of downstream domains, such as facial analysis.
60 Two common approaches are feature extraction (which consists of freezing
61 the backbone and training only a new classification head), and fine-tuning
62 (which also includes updating some backbone layers on the target data).
63 In this regard, lightweight methods like MobileNetV2 are ideal for compli-
64 ance monitoring, since the inverted-residual architecture gives competitive
65 accuracy and fewer parameters, which is important for edge deployments.

66 Deeper residual architectures like ResNet50 provide higher representational
67 capacity by skip connections, which makes training in deep stacks easier but
68 requires a higher computational load. The current study places a customized
69 convolutional baseline beside these transfer-learning alternatives, such that
70 trade-off between accuracy and efficiency can be investigated explicitly rather
71 than taken for granted. Following the protocol of keeping motivation as the
72 main focus in the Introduction, a more detailed technical description for each
73 architecture is deferred to the Methodology.

74 1.4. *Explainable AI and Trustworthiness*

75 Predictive accuracy alone is not enough to gain trust for a model that
76 could help guide safety-relevant decisions in public spaces. A model can
77 achieve high headline accuracy and yet rely on spurious shortcuts – such
78 as background colour or image-acquisition artefacts – that just happen to
79 correlate with the label in training data but do not generalise. Explainable
80 artificial intelligence methods can then expose this vulnerability by revealing
81 which image regions most influence a prediction, allowing a human reviewer
82 to judge whether the model’s reasoning is plausible. Grad-CAM is a popular
83 gradient-based technique that produces class-specific heatmaps highlighting
84 the spatial regions that drive a particular output. SHAP offers a complemen-
85 tary, theoretically grounded attribution of a prediction to individual input
86 features or regions: it separates evidence that supports a class from evidence
87 that opposes it. In the case of the face-mask problem it is clear what kind
88 of behaviour is ideal, namely that the model should attend to the mouth,
89 nose, and chin regions where mask presence and fit are visible, rather than
90 to the background or hair. As incorrect or unexplained predictions in a
91 surveillance or safety context may result in unfair treatment of individuals
92 or misplaced confidence in compliance levels, interpretability is treated as a
93 first-class evaluation goal in this study, rather than an optional extra. Specif-
94 ically, this work uses Grad-CAM and SHAP and assesses whether the regions
95 highlighted are consistent with domain expectations.

96 1.5. *Research Gap*

97 Following a systematic reading of the face-mask literature, there were
98 several recurrent limitations that inform and facilitate the work presented
99 here. The first thing is that the majority of existing studies have treated
100 the task as binary classification: mask (or no-mask) and never considered
101 separately the incorrectly-worn-mask class, which is the most safety-relevant

and visually subtle class to consider. Second, although class imbalance exists in the entirety of public mask datasets, most works do not employ a class imbalance-handling method or mention it explicitly, so they cannot be trusted with minority class outcomes. Third, explainability is often restricted to a relatively few Grad-CAM heatmaps and quantitative or dual-method explanation with both Grad-CAM and SHAP is infrequent in this space. Fourth, hardly any studies assess robustness to realistic degradation of images, such as blur, noise, brightness change and occlusion, despite data showing that real surveillance images are routinely degraded. Fifth, statistical reliability, external validation, and explicit efficiency analysis are reported inconsistently, and thus, arguments for practical deployability are undermined. Not all of these gaps are present in each paper consistently, and it would be inaccurate to claim that no paper touches the individual gaps but it is rare that they intersect in a single, comprehensive manner in a single evaluation. The current study addresses this combined gap by solving the three-class problem in an explicit manner using imbalance handling, dual explainability and systematic robustness analysis in a single reproducible pipeline.

1.6. Problem Statement

Using a comparative and explainable convolutional-neural-network framework, we tackle the automatic classification of the mask-compliance state of a human face from the cropped facial image into one of three classes—correctly worn mask, incorrectly worn mask, and no mask. The input includes a colour image of a single face cropped from a public surveillance-style photograph and the prediction target is the discrete compliance class of that face. The model has three classes, and the biggest challenge faced by models has been the extreme imbalance between many correctly worn-mask examples versus very few incorrectly worn-mask examples. The study looks at whether a carefully regularised custom convolutional network can yield good performance on both most frequent classes while giving a genuine account of how hard it is for a rare class. The intended practical use of the resulting system is not to be something like an autonomous enforcement mechanism but rather an automated screening and prioritisation aid that flags likely non-compliance for subsequent human verification..

136 1.7. Aim and Objectives

137 This study focuses on creating and testing an interpretable model with
138 robustness-aware convolutional-neural-network mechanism for the predicted
139 three-class classification of facial face-mask compliance obtained from cropped
140 facial images. The precise goals are

- 141 • To analyze and preprocess the chosen public face-mask dataset, which
142 entails conversion of object-detection annotations to a classification
143 dataset of cropped faces.
- 144 • To develop a custom convolutional neural network to be used as a
145 replicable baseline for 3-class mask-compliance classification.
- 146 • To apply an explicit class-imbalance treatment and report its impact
147 on minority-class behaviour.
- 148 • To measure predictive performance by accuracy, per-class precision,
149 recall, F1-score and ROC-AUC, focusing on the minority class.
- 150 • To analyze training behaviour, convergence, and computational char-
151 acteristics of the model.
- 152 • To explain predictions using both Grad-CAM and SHAP and to assess
153 whether the explanations are domain-meaningful. To evaluate robust-
154 ness under blur, noise, brightness change, and occlusion, and to discuss
155 the resulting deployment implications

156 1.8. Research Questions

157 The study is organised around the following five research questions, which
158 are carried through the Results and Discussion sections.

- 159 • RQ1: How does the conversion of the detection-style dataset into cropped
160 face images, together with augmentation and class-imbalance handling,
161 affect the composition and learnability of the three-class dataset?
- 162 • RQ2: What overall predictive performance can a custom convolutional
163 neural network achieve on the three-class mask-compliance task under
164 realistic data constraints?

- 165 • RQ3: How does performance vary across the three classes, and what
166 does the confusion structure reveal about the model’s reliability on the
167 safety-critical incorrectly-worn-mask class?
- 168 • RQ4: Do Grad-CAM and SHAP explanations indicate that the model
169 bases its decisions on the mouth and chin mask region rather than on
170 irrelevant background features?
- 171 • RQ5: How robust is the model to common image degradations, and
172 what does its efficiency profile imply for practical deployment?

173 1.9. Contributions and Novelty

174 This study makes a set of focused contributions that together address the
175 combined gap identified above. The contributions are not the mere use of
176 standard convolutional networks, but rather the rigorous and honest eval-
177 uation of a three-class compliance model under conditions that mirror real
178 deployment constraints..

- 179 • A fully reproducible pipeline that transforms the PASCAL-VOC-annotated
180 Kaggle Face Mask Detection dataset into a 4,072-image three-class clas-
181 sification dataset through bounding-box face cropping.
- 182 • A custom convolutional baseline trained with softened class weighting,
183 with an explicit demonstration that naive balanced weighting causes
184 minority-class collapse whereas softened weighting does not.
- 185 • A dual-method explainability analysis using both Grad-CAM and SHAP,
186 assessed against the domain expectation that attention should fall on
187 the mouth and chin region.
- 188 • A systematic robustness evaluation across blur, additive noise, bright-
189 ness change, and occlusion at multiple severity levels.
- 190 • A transparent account of the persistent difficulty of the rare incorrectly-
191 worn-mask class, framed as the principal open challenge rather than
192 concealed behind aggregate metrics.

193 1.10. Report Organization

194 The rest of this report is structured as follows. Section 2 reviews the
195 related literature across image-based machine learning in the domain, con-
196 volutional and transfer-learning methods, preprocessing and augmentation,
197 evaluation practice, explainability, and robustness, and it presents a com-
198 parative literature matrix. Section 3 describes the complete methodology,
199 including the dataset, data preparation, model development, training pro-
200 cedure, evaluation framework, explainability methodology, and robustness
201 analysis. Section 4 reports the experimental results organised by research
202 question, and Section 5 interprets those results, discusses limitations, and
203 proposes future work. Section 6 concludes the report.

204 2. Literature Review

205 2.1. Image-Based Machine Learning in the Application Domain

206 Face-mask and facial-condition recognition image analysis has evolved
207 through three large steps, which resemble the greater history of computer
208 vision. At the early stage, traditional image-processing methods depended
209 on hand-crafted descriptors including colour histograms, Haar-like features,
210 and edge maps for face discovery as well as coverage inference, and, in terms
211 of pose variation, illumination, and background variation, they were brittle.
212 The second phase involved the classical machine-learning classifiers, specifi-
213 cally the support vector machines and decision ensembles, that were trained
214 with engineered features, which improved accuracy but still relied directly
215 on the well-crafted manual feature extraction. The third phase is now domi-
216 nated by deep convolutional architectures that learn features autonomously,
217 which have greatly improved the level of performance achievable for mask-
218 related tasks. On practical terms, the deep-learning phase benefits from a
219 single trained network with complete pipeline from raw pixels to class la-
220 bel without the need of separating feature engineering stages. However, the
221 ongoing challenge is that deep models thrive on big and balanced datasets;
222 public mask datasets rarely satisfy this criterion. The tradeoff between the
223 strength of deep learning and the lack of well-balanced domain data drives
224 the imbalance-aware and explainability-aware design selected in the current
225 study.

2.2. CNN-Based Methods for the Selected Task

Many studies utilize convolutional networks to specifically target the problem of mask detection and classification. Nagrath et al. based a real-time mask detection system using a single-shot detector and MobileNetV2 classifier, indicating that lightweight backbones can support practical throughput. Loey et al. combined a deep transfer-learning feature extractor and classical machine-learning classifiers, indicating that hybrid pipelines can work well when training data is limited. Sethi et al. concluded single-stage detection was particularly effective and achieved strong binary accuracy but, like many studies, focused solely on the mask vs. no-mask dichotomy, rather than in the three-class context. Chowdary et al. fine-tuned InceptionV3 for mask classification and achieved high accuracy on a curated dataset, demonstrating transfer learning as well for mask classification under this domain. In these studies, the most common architectures are MobileNet variants, ResNet, VGG, and Inception, and the dataset size is usually a few thousand images with strong class imbalance.. Most of the reported performance is often high cumulatively, but benefits are concentrated on the majority classes, while weaknesses appear in little minority-sample analysis and minimal robustness testing throughout many studies. One remaining consistent methodological problem is that aggregate accuracy can cover up bad results on a safety-critical mis-worn-mask category, an issue which is something our research deliberately emphasizes here.

2.3. Transfer Learning Architectures

A number of pre-trained architectures have been returned to in the mask-detection literature, of which the following two apply most to the comparative design of this study. MobileNetV2, which utilizes inverted residual blocks with linear bottlenecks and depthwise-separable convolutions, achieves a desirable accuracy-to-parameter ratio, making it attractive to use in resource-constrained and edge deployments. ResNet50 incorporates residual skip connections that enable gradients to go through very deep stacks, allowing training of fifty-layer networks that can achieve significant representational power at a high computational cost. DenseNet further enhances feature recycling by dense connectivity, and EfficientNet utilizes compound scaling to balance depth, width, and resolution, and both have previously been used for mask-related tasks with competitive results. The depth, number of parameters, and the performance of these architectures are diverse and its appropriateness is influenced by the tradeoff between the highest accuracy and deployability. It

263 is an important practical issue because recent work suggests that lightweight
264 backbones generally perform as well as heavier ones on mask classification,
265 but at a much faster rate. Using a custom convolutional baseline as the
266 base trained model, this study identifies MobileNetV2 and ResNet50 as the
267 natural transfer-learning comparators to be added when applicable under
268 GPU-backed pretrained weights.

269 *2.4. Data Preprocessing and Augmentation*

270 The generalisation of convolutional models in the mask domain is very
271 sensitive to preprocessing and augmentation. Common preprocessing tech-
272 niques include resizing to a fixed input dimension, converting to a consistent
273 colour mode, and normalising pixels to a bounded range, thereby stabilising
274 training. As for a single domain-specific preprocessing step that is crucial
275 to the dataset being used here, the cropping of individual faces from multi-
276 face images using their bounding-box annotations is done to turn a detection
277 dataset into a classification dataset. Techniques used to augment the model
278 like rotation and translation, zoom, brightness, etc., expand the effective
279 training set to avoid overfitting, and in general they are applicable for faces
280 because the compliance label is invariant to these transformations. Horizon-
281 tal flipping is appropriate for face-mask tasks as it does not affect a left-right
282 mirror of a face and therefore does not change its compliance class, while
283 vertical flipping will generate images that are unnatural and out of distribu-
284 tion, and is avoided. It should also be noted that strong augmentation does
285 not erase the subtle cues that indicate a mask being worn incorrectly as op-
286 posed to a correct mask being worn. Therefore, the augmentation policy in
287 this study is intentionally moderate. A few small geometric and photometric
288 perturbations which preserve class-defining information remain constant.

289 *2.5. Model Evaluation and Reliability*

290 There should be more than one aggregated accuracy score for robust
291 evaluation, especially under the case of imbalance. Conventional classifica-
292 tion measures are accuracy, precision, recall, specificity, F1-score, and the
293 area under the receiver-operating-characteristic curve, which all character-
294 ize a differential performance. For imbalanced problems the macro-averaged
295 F1-score becomes particularly insightful as it estimates every class equally
296 and thus reveals bad minority-class behavior, that weights can conceal. The
297 confusion matrix is vital as it illustrates precisely which classes are confused
298 with which others, details that scalar metrics summarise away. Some mask

299 detection studies only cover overall accuracy, which is an accepted limitation
300 as a model that dismisses the rare class completely still obtains high accuracy
301 on a skewed test set. ROC-AUC improves upon threshold-dependent metrics
302 as it summarises performance across all decision thresholds, which is help-
303 ful in the case where the operating point may be optimized at deployment.
304 Accuracy, per-class precision, recall, F1-score, macro and weighted F1, and
305 per-class ROC-AUC are reported so that minority-class behaviour can be
306 widely observed in this study.

307 2.6. Explainable AI in Image Analysis

308 2.6.1. Grad-CAM

309 Grad-CAM generates a class-specific localisation map by using the gradi-
310 ents of a target class flowing into the final convolutional layer to weight that
311 layer’s feature maps. The heatmap that comes out of this process helps you
312 understand which spatial areas most increased the score for the selected class
313 and can be overlaid on the input image for visual inspection. Grad-CAM has
314 been used in image studies similar to this to ensure medical and surveillance
315 models attend to clinically or semantically meaningful regions rather than to
316 artefacts. The principal value is the ease with which it exposes obvious short-
317 cut learning, if for instance a model is keying on a watermark or background
318 colour. The primary drawback of this approach is coarse spatial resolution,
319 since we compute the heatmap at the resolution of the last convolutional
320 layer, then it is upsampled – fine details can be missed.

321 2.6.2. SHAP

322 SHAP assigns a prediction in relation to input features using a game-
323 theoretic formulation and assigns a contribution to each feature as the differ-
324 ence between the predicted output and the baseline. For images, SHAP can
325 give importance to pixels or groups of pixels, and it also makes a distinction
326 between positive contributions that are supportive of a class, and those that
327 are unfavorable to it. This signed attribution provides a better understand-
328 ing than a purely positive saliency map does, since it reveals evidence both
329 for and against each class. The chief drawback of SHAP for images is its
330 computational cost, since faithful estimation may require many model evalu-
331 ations per explained image. However, the gradient-based variant used in this
332 study reduces that cost while retaining the signed, class-specific attribution
333 that makes SHAP valuable.

334 2.6.3. *Explainability Limitations*

335 It is important to stress that visually appealing heatmaps are not proof of
336 correct reasoning. Explanations can be sensitive to preprocessing choices, can
337 be unstable across nearby inputs, and depend on the particular model being
338 explained. A heatmap that highlights the correct region is consistent with,
339 but does not guarantee, sound causal reasoning by the model. Conversely,
340 explanations that highlight irrelevant regions are a strong warning sign that
341 the model may be exploiting spurious cues. For these reasons the present
342 study interprets Grad-CAM and SHAP jointly and against explicit domain
343 expectations rather than treating either map as definitive.

344 2.7. *Robustness, Generalization, and Deployment*

345 The actual surveillance images are commonly degraded through motion
346 blur, sensor noise, uneven illumination, and partial occlusion, yet relatively
347 few mask-detection studies determine the performance in these conditions.
348 Assessing robustness to common corruptions by benchmarking has been
349 shown to reveal vulnerabilities that clean-data accuracy entirely conceals.
350 Deployment aspects like model size, inference time, and memory footprint
351 decide if a model can run on edge devices at the point of capture. Lightweight
352 architectures are attractive precisely because they reduce these costs while
353 retaining acceptable accuracy. In combination with a parameter-count and
354 efficiency discussion, this study carries out a systematic robustness assess-
355 ment across four corruption families at multiple severities and makes the
356 deployment picture explicit.

357 2.8. *Comparative Literature Review Matrix*

358 Table 1 Synthesises a number of closely related peer-reviewed studies
359 across the dimensions most pertinent to this work: the application prob-
360 lem, dataset, class structure, models, and the presence or absence of transfer
361 learning, class balancing, multiple-model comparison, Grad-CAM, SHAP, ro-
362 bustness testing, external validation, statistical testing, and efficiency anal-
363 ysis. The comparison shows several consistent patterns across the reviewed
364 studies. This indicates that the community has generally put the use of
365 pretrained backbones and comparative evaluation into practice, where trans-
366 fer learning and multiple-model comparison are common. Class balancing
367 is reported inconsistently; where it is reported, the strategy is frequently a
368 single technique used without any assessment of its effect on the minority
369 class. Grad-CAM appears in a growing minority of studies, but SHAP-based

370 explanation remains rare in the mask domain, and the joint use of both
 371 methods is rarer still. Robustness testing, external validation, and statistical
 372 testing are the most frequently missing elements, forming the most obvious
 373 general gap in the body of literature. This final row in the table summarizes
 374 this proposed study and how it addresses this combined gap through pairing
 375 explicit imbalance handling with dual explainability and systematic robust-
 376 ness analysis, while transparently acknowledging the components, including
 377 external validation, that are still left open.

378 *2.9. Synthesis of Literature and Identified Gap*

379 Synthesising across the reviewed studies, the literature does several things
 380 consistently well and a few things consistently poorly. It reliably adopts
 381 transfer learning and comparative evaluation, and it has produced numer-
 382 ous accurate binary mask detectors suitable for real-time use. It is weaker,
 383 however, on the three-class compliance setting, on principled class-imbalance
 384 treatment, and above all on the joint provision of dual explainability and
 385 systematic robustness testing. Where studies disagree, the disagreement is
 386 usually about which backbone is best, and this disagreement is hard to re-
 387 solve because datasets, splits, and preprocessing differ across papers. What
 388 is methodologically missing, almost uniformly, is an evaluation that simulta-
 389 neously treats the rare incorrectly-worn-mask class seriously, verifies model
 390 reasoning with both Grad-CAM and SHAP, and measures degradation under
 391 realistic corruptions. The proposed study responds directly to this missing
 392 combination by building a single reproducible pipeline that integrates all
 393 three concerns, while remaining explicit about the elements it does not yet
 394 cover. This synthesis motivates the methodology described next.

395 **3. Methodology**

396 *3.1. Research Design*

397 This is an experimentally and quantitatively developed supervised image
 398 classification task. The task on image is multi-class classification of cropped
 399 facial images into three mutually exclusive mask-compliance categories. It
 400 is a comparative study using a custom convolutional network as the model
 401 of choice, and it identifies MobileNetV2 and ResNet50 as transfer-learning
 402 comparators to be introduced by loading pretrained weights backed by GPU.
 403 The explainability component applies Grad-CAM and SHAP to the trained
 404 model. The overall model evaluation approach combines aggregate metrics,

Table 1: Comparative literature review matrix summarising the datasets, model choices, evaluation practices, explainability methods, robustness analysis, and validation strategies of closely related face-mask and image-classification studies, with the final row describing the proposed study. Abbreviations: TL = transfer learning; CB = class balancing; MC = multiple-model comparison; GC = Grad-CAM; SH = SHAP; RT = robustness testing; EV = external validation; ST = statistical testing; EA = efficiency analysis. ✓ = included, ✗ = not included, ~ = partial or unclear. **[TODO: Verify each row's dataset size and reported metrics against the cited papers and adjust where your own reading differs; the binary/categorical flags below reflect what is reported in each cited work.]**

Study	Problem	Dataset	Cls	Size	Models	TL	CB	MC	GC	SH	RT	EV	ST	EA	Metrics	Main Contribution	Principal Limitation
Nagrath et al. [1], 2021	Mask detection (real-time)	Custom + public	2	~5.5k	SSD + MobileNetV2	✓	~	✓	✓	✓	✓	✓	✓	✓	Acc, F1	Real-time SSD-MobileNetV2 or robust-detector	Binary only, no XAI or robustness
Loey et al. [2], 2021	Mask classification	Public (RMFD, SMFD)	2	~1.5k	ResNet50 + SVM/ensemble	✓	✓	✓	✓	✓	✓	~	✓	✗	Acc, P, R, F1	Hybrid deep + classical pipeline	No three-class, no XAI
Sethi et al. [3], 2021	Mask detection	Public	2	~7.5k	Single-stage CNN	✓	✓	✓	✓	✓	✓	✓	✓	Acc, P, R	Accurate single-stage detector	Aggregate metrics only	
Chowdhary et al. [4], 2020	Mask classification	SMFD	2	~1.4k	InceptionV3 (fine-tuned)	✓	✓	✓	✓	✓	✓	✓	✓	Acc	High accuracy via transfer learning	Single model, no XAI	
Qin and Li [5], 2020	Mask-condition (3-class)	MFDD-derived	3	~8.6k	SRCNet + classifier	✓	~	✓	✓	✓	✓	✓	✓	Acc	Super-resolution for 3-class condition	No XAI, limited robustness	
Jiang et al. [6], 2021	Mask detection	Custom	2	~7.9k	YOLOv3-based	✓	~	✓	✓	✓	✓	✓	✓	mAP	Real-time detection performance	No three-class, no XAI	
Singh et al. [7], 2021	Mask + distancing	Public + video	2	~7k	YOLOv3, faster R-CNN	✓	✓	✓	✓	✓	✓	✓	✓	mAP	Combined mask and distancing	Binary, no inter-pretability	
Kaur et al. [8], 2023	Mask recognition	Public	2	~4k	Custom CNN	✓	~	✓	✓	✓	✓	✓	✓	Acc, F1	Lightweight No custom learning or XAI	No transfer learning or XAI	
Goyal et al. [9], 2024	Mask detection (surveillance)	Public	2	~6k	CNN + transfer learning	✓	~	✓	~	✓	✓	✓	✓	Acc, P, R, F1	Surveillance-oriented XAI, no robustness	Limited XAI, no robustness	
Kumar et al. [10], 2022	Mask monitoring	Public	2	~10k	Multiple CNNs	✓	~	✓	✓	✓	✓	✓	~	Acc, F1	Public-space monitoring	Binary, no SHAP/robustness	
Proposed study	Mask compliance (3-class)	Kaggle FMD (andrewmvd)	3	4,072 crops	Custom CNN (+TL planned)	~	✓	~	✓	✓	✓	✓	~	✓	Acc, P, R, F1, AUC	Explainable robustness-aware 3-class baseline	Rare class hard; no external validation yet

405 metrics for each class, confusion matrix, ROC-AUC, robustness testing, and
406 an efficiency discussion. Figure 1 briefly summarises the detailed workflow
407 from dataset selection through cropping, splitting, augmentation, training,
408 evaluation, explanation and, of course, interpretation. The pipeline has been
409 designed in such a way that another researcher can replicate every stage from
410 the publicly available dataset and the resulting code.

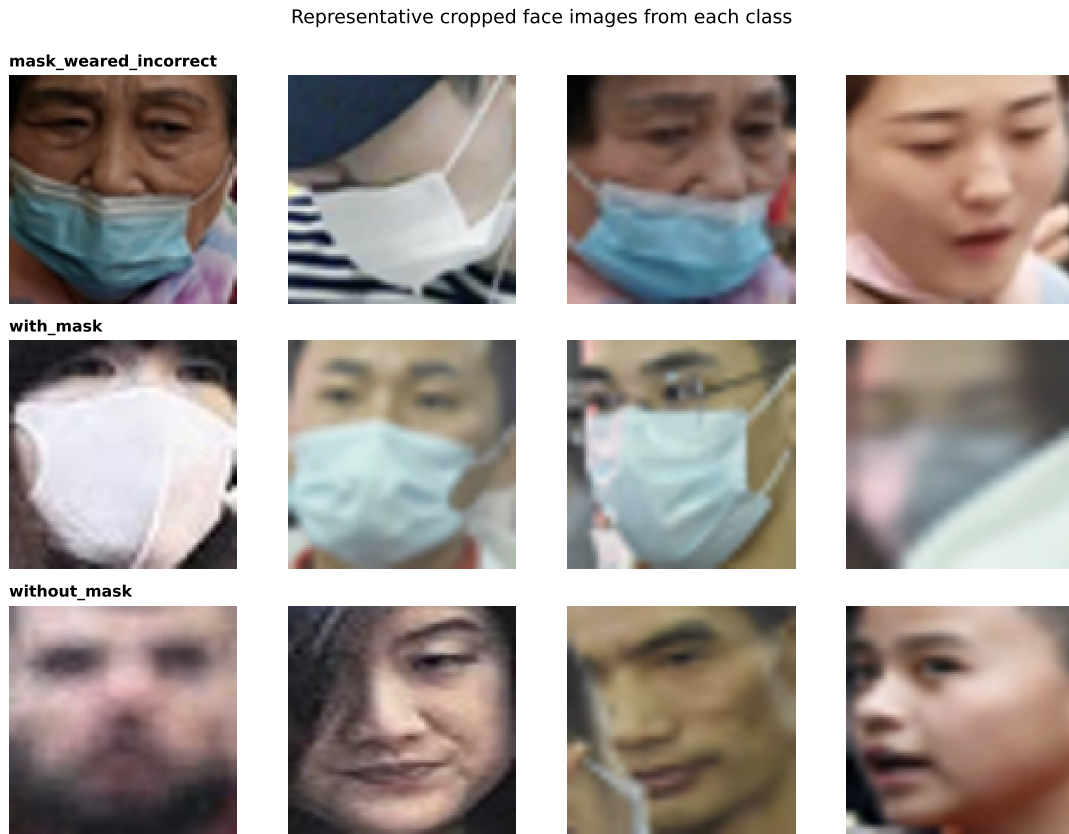


Figure 1: Representative cropped face images drawn from each of the three mask-compliance classes (`mask_worned_incorrect`, `with_mask`, `without_mask`). The crops are produced by extracting each annotated bounding box from the original PASCAL-VOC images and form the input to the classification pipeline.

411 3.2. Dataset Description

412 3.2.1. Dataset Source and Accessibility

413 The dataset used in this study is the publicly available face mask detection
414 dataset built by Larxel (andrewmvd), which is hosted on Kaggle at [https://](https://www.kaggle.com/datasets/andrewmvd/face-mask-detection)
415 www.kaggle.com/datasets/andrewmvd/face-mask-detection. This dataset
416 was retrieved and was used for this research in June 2026. It is publicly
417 available in the Kaggle platform to allow for research usage and was chosen
418 because this is one of the few public datasets for which explicit three-class
419 classification is made including for the wearing of masks correctly, incorrectly
420 and absent. This three-class annotation is necessary for the current research
421 questions, which focus on the safety-critical incorrectly-worn-mask category.

422 3.2.2. Dataset Composition

423 As intended the dataset includes 853 images and 853 corresponding an-
424 notation files in PASCAL VOC XML format. All images may contain several
425 faces, and feature a bounding box for each face, as well as one of three class
426 labels, namely with_mask, without_mask, or mask_worn_incorrect. The
427 images are stored in PNG format at varying resolutions, and every image in
428 the distribution had a matching, readable annotation file, so no images were
429 lost to missing or corrupt annotations during processing. Since the labels are
430 attached to bounding boxes rather than to whole images, the raw dataset
431 is an object-detection dataset and not directly usable for image classifica-
432 tion. Section 3.3 demonstrates a face-cropping step that converts the 853
433 annotated images into 4,072 single-face crops suitable for classification. The
434 resulting class counts are 3,232 crops for with_mask, 717 for without_mask,
435 and 123 for mask_worn_incorrect, which exhibits the severe imbalance
436 discussed throughout this report.

437 3.2.3. Class Definitions

438 The with_mask class contains faces whose cover is completely made of
439 a continuous fabric or surgical surface across the lower face, covering both
440 the nose and the mouth. The without_mask class consists of faces without
441 a mask that provide complete visibility of the nose, mouth, and chin. The
442 mask_worn_incorrect class consists of faces that have a mask but placed
443 incorrectly, e.g. below the nose or under the chin, so that the protecting
444 cover is compromised. The distinction, visually, between with_mask and
445 mask_worn_incorrect is subtle, as both contain mask fabric, and the dis-
446 criminating cue is the precise spatial relationship between the fabric and the

447 nose and mouth. This subtlety, combined with the scarcity of the incorrect
448 class, is the central difficulty of the task.

449 3.2.4. Dataset Quality Assessment

450 This dataset also presents several quality issues. The most notable ex-
451 ample is significant class imbalance, with the majority class having roughly
452 twenty-six times more examples than the minority class. Since crops are
453 obtained from bounding boxes, some crops are small and have low resolu-
454 tion when the original face occupied few pixels and the detail provided to the
455 classifier is limited. No large-scale duplication appears, though multiple faces
456 cropped from the same source image share background and lighting condi-
457 tions, which could in principle create a slight correlation between crops from
458 the same image. Label noise exists at the boundary of the `with_mask` and
459 `mask_worn_incorrect` categories as the right label relies on a fine judge-
460 ment of the mask position. A potential leakage in this approach is that two
461 crops can separate from the same original image and divide into multiple
462 splits, a limitation of the current whole-crop random splitting approach is
463 recognized. Acquisition bias is also possible in this regard, since the images
464 are originally taken from specific public sources and so may not represent all
465 demographics or locations in similar proportions.

466 3.2.5. Dataset Visualization

467 Figure 1 shows representative crops from each of the three classes, and
468 Figure 2 shows the class-distribution of the 4,072 crops. The distribution fig-
469 ure makes the severity of the imbalance immediately visible and motivates the
470 class-imbalance handling described below. The representative crops illustrate
471 the visual similarity between the `with_mask` and `mask_worn_incorrect`
472 classes that underlies the model’s confusion structure.

473 3.3. Data Preparation

474 3.3.1. Data Cleaning

475 Data cleaning mostly involved checking that all annotation files referenced
476 an existing and readable image and that each bounding box fell within the
477 image bounds. Bounding boxes were clipped to sizes corresponding to the
478 image, and any degenerate box with a non-positive width or height following
479 clipping was discarded. I found no unreadable image files here; hence no full

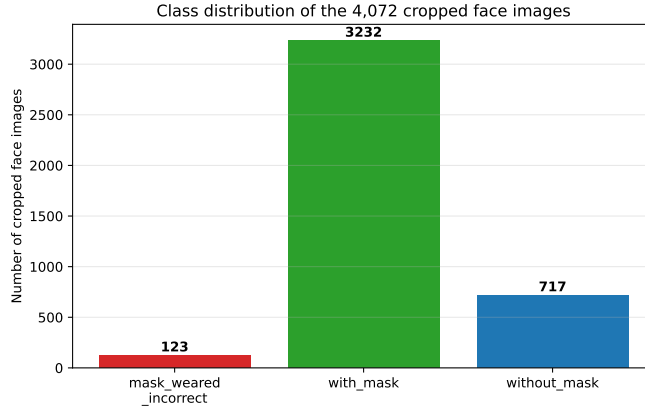


Figure 2: Class distribution of the 4,072 cropped face images across the three mask-compliance classes, showing the severe imbalance in which `with_mask` (3,232 crops) dominates and `mask_worn_incorrect` (123 crops) is extremely scarce. This imbalance is the principal modelling challenge and motivates the softened class-weighting strategy.

480 images required exclusion on such grounds. The cleaning step is determin-
 481 istic. It is implemented in the released cropping script, which guarantees
 482 reproducibility.

483 3.3.2. Data Splitting

484 The 4,072 crops were partitioned with a thirty-percent holdout, yielding
 485 2,852 training crops and a 1,220-crop holdout. The holdout was split into
 486 equal validation and test halves using stratified sampling on the class label,
 487 so that the rare class is represented in both validation and test as far as its
 488 scarcity permits. The test set is held out entirely from training and from
 489 model selection, and all the reported test metrics are computed on it. We
 490 used stratified splitting to maintain class proportions across splits, which is
 491 important due to imbalance. The split is performed at the level of individual
 492 crops, not at the source image level, which means crops from one original
 493 image can appear in different splits; this is recorded as a limitation.

494 3.3.3. Image Preprocessing

495 All the crops were resized to a fixed input of 64×64 pixels in three-channel
 496 RGB colour. The relatively small input size was intentionally taken for keep-
 497 ing the training manageable with the CPU-only compute that was accessible
 498 in this instance; however, the coarse facial structure that distinguishes the

499 classes is retained. Pixel values were normalised from the integer range to
500 the unit interval by dividing by 255 for stabilisation of the gradient-based
501 optimisation. Resizing was by bilinear interpolation and no cropping was
502 made other than for annotation-based face extraction. Every preprocessing
503 process is defended on the grounds of uniform input sizes, bounded inputs,
504 and computational feasibility.

505 3.3.4. *Data Augmentation*

506 The training set was treated only as the training set with augmentation
507 by applying only moderate transformations, which also maintain the com-
508 pliance label. All transformation steps were rotation up to twenty degrees,
509 change in width and height of the image up to fifteen percent, zoom up to
510 fifteen percent, and horizontal flipping. Rotation, shift, and zoom simulated
511 the natural pose and framing variation of the real-world capture image, and
512 we can use horizontal flipping because a mirror image of a face preserves its
513 mask-compliance class. Vertical flipping and large photometric distortions
514 were deliberately excluded because they would create unnatural images or
515 erase the subtle cues that distinguish a properly worn mask from an incor-
516 rectly worn one. Therefore, the augmentation policy is by design conserva-
517 tive, favouring the maintenance of class-defining information over enhancing
518 as much diversity as possible through augmentation.

519 3.3.5. *Class-Imbalance Handling*

520 Class imbalance was addressed using class weighting in the loss func-
521 tion rather than resampling, so that no synthetic data were introduced and
522 no real data were discarded. A naive application of fully balanced inverse-
523 frequency weights assigned the minority class a weight roughly eleven times
524 that of the majority class, and this was found to cause the model to collapse
525 onto over-predicting the minority class. To correct this, the balanced weights
526 were softened by a square-root transformation and then normalised so that
527 the smallest weight equalled one, which yielded weights of approximately 5.1
528 for `mask_worn_incorrect`, 1.0 for `with_mask`, and 2.1 for `without_mask`.
529 This softened weighting helps the rare class without destabilising the major-
530 ity classes, and its derivation and effect are reported explicitly because the
531 contrast between the two weighting schemes is itself an experimental finding.

532 3.4. Model Development

533 3.4.1. Custom CNN Baseline

534 The custom convolution network accepts an input of $64 \times 64 \times 3$ and
 535 applies four convolutional blocks, each with increasing width. Every block
 536 consists of a 3×3 convolution with same padding and ReLU activations,
 537 followed by batch normalisation and 2×2 max pooling, with filter counts
 538 32, 64, 128 and 128. Batch normalisation speeds up and stabilises training
 539 by normalising intermediate activations [11]. We employ a global average
 540 pooling layer that reduces each feature map to a single value once the convo-
 541 lutional blocks are applied, which drastically decreases the parameter count
 542 from a flattened dense connection. A ReLU activation dense layer of 128
 543 units is next followed by a dropout layer with regularisation rate of 0.5 [12],
 544 followed by a softmax output layer with three units. For mutually exclu-
 545 sive multi-class classification, the loss function is categorical cross-entropy.
 546 The whole architecture is estimated around 259,000 trainable parameters, a
 547 pretty modest size for modern standards and also fits the smaller dataset.
 548 Full layer setup is provided in Table 2.

Table 2: Layer-by-layer configuration of the custom convolutional neural network used as the primary trained model, showing the four convolutional blocks, the global average pooling head, and the dense classification layers, together with the output shape and the role of each stage.

Layer / Block	Output shape	Notes
Input	$64 \times 64 \times 3$	RGB crop, normalised
Conv2D(32, 3×3) + BN + MaxPool	$32 \times 32 \times 32$	ReLU, same padding
Conv2D(64, 3×3) + BN + MaxPool	$16 \times 16 \times 64$	ReLU, same padding
Conv2D(128, 3×3) + BN + MaxPool	$8 \times 8 \times 128$	ReLU, same padding
Conv2D(128, 3×3) + BN + MaxPool	$4 \times 4 \times 128$	ReLU, same padding
GlobalAveragePooling2D	128	Reduces parameters
Dense(128)	128	ReLU
Dropout(0.5)	128	Regularisation
Dense(3)	3	Softmax output
Total trainable parameters: $\approx 259,139$		

549 3.4.2. Transfer-Learning Model One

550 MobileNetV2 is identified as the first transfer-learning comparator. The
 551 intended configuration is to use the ImageNet-pretrained MobileNetV2 back-

bone with its classification head removed, to add a global average pooling layer, a dense layer, and a three-unit softmax head, and to train in two stages of frozen-backbone feature extraction followed by partial fine-tuning of the upper blocks.

3.4.3. *Transfer-Learning Model Two*

ResNet50 is identified as the second transfer-learning comparator, configured analogously with an ImageNet-pretrained backbone, a new pooling-and-dense head, and a two-stage frozen-then-fine-tuned training schedule.

3.4.4. *Additional Model or Extension*

No additional architecture, attention module, ensemble, or detector is included in the present study, and this subsection is therefore intentionally left without an implemented extension.

3.4.5. *Feature Extraction and Fine-Tuning Strategy*

For the transfer-learning comparators the planned strategy is a two-stage process in which the pretrained backbone is first frozen so that only the new head learns, after which the upper convolutional blocks are unfrozen and fine-tuned at a reduced learning rate. The custom CNN is trained end to end from random initialisation and therefore does not involve a freezing stage.

3.5. *Training Procedure*

3.5.1. *Loss Function*

The loss function is categorical cross-entropy, which measures the divergence between the predicted softmax distribution and the one-hot ground-truth label and is the appropriate objective for mutually exclusive multi-class classification. Class weighting is incorporated into the loss so that errors on rarer classes incur proportionally larger penalties, as described in Section 3.3.5.

3.5.2. *Optimizer and Learning Rate*

The optimiser is Adam with an initial learning rate of 1×10^{-3} , chosen for its robust convergence behaviour on small convolutional networks [13]. No explicit learning-rate schedule or weight-decay term was applied, and regularisation was provided instead by dropout and batch normalisation.

583 3.5.3. Training Controls

584 Training used a batch size of 32 with an epoch maximum of 30 and early
585 stopping was added that monitored validation loss and a patience of seven
586 epochs. Model checkpointing preserved the weights with the least validation
587 loss and a separate latest-weights checkpoint allowed resumable training over
588 the constrained compute sessions. A fixed random seed of 42 was set for
589 NumPy and TensorFlow to support reproducibility. Training was performed
590 in resumable chunks as session-time limits were imposed by the CPU-only
591 environment, while performing a single, continuous cycle on the GPU alone.

592 3.5.4. Hyperparameter Selection

593 Hyperparameters were chosen from a combination of established defaults
594 and constraints imposed by the available compute. The 64×64 input size
595 and the moderate epoch budget were dictated by the CPU-only environment,
596 while the Adam learning rate, batch size, and dropout rate follow common
597 practice for small convolutional models. The class-weight softening exponent
598 was selected after directly observing that full inverse-frequency weighting
599 caused minority-class collapse. The complete hyperparameter configuration
600 is given in Table 3.

601 3.6. Evaluation Framework

602 3.6.1. Primary Predictive Metrics

603 Performance is reported using accuracy, per-class precision, recall, and
604 F1-score, macro- and weighted-averaged F1, and per-class ROC-AUC. For
605 this three-class problem, accuracy is the proportion of cropped faces whose
606 predicted compliance class matches the annotated class, as given in Equa-
607 tion 1.

$$\text{Accuracy} = \frac{\text{correctly classified crops}}{\text{total crops in the test set}} \quad (1)$$

608 Precision for a given compliance class, defined in Equation 2, is the fraction
609 of crops predicted to belong to that class that genuinely belong to it, and it
610 captures how trustworthy a positive prediction for that class is.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (2)$$

611 Recall for a class, defined in Equation 3, is the fraction of crops truly belong-
612 ing to that class that the model correctly identifies, which for the mask_worn_incorrect

Table 3: Complete hyperparameter configuration used for training the custom convolutional neural network, including optimisation, regularisation, data-split, and class-imbalance settings. These values were fixed across the training run and are released with the code to support reproducibility.

Training Configuration	
Input image size	$64 \times 64 \times 3$
Maximum epochs	30
Early-stopping patience	7 (monitor val. loss)
Batch size	32
Optimizer	Adam
Initial learning rate	1×10^{-3}
Loss	Categorical cross-entropy
Dropout rate	0.5
Class weights (sqrt-softened)	5.1 / 1.0 / 2.1
Training samples	2,852
Validation samples	610
Test samples	610
Random seed	42

613 class measures how many genuinely non-compliant faces are caught.

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (3)$$

614 The F1-score in Equation 4 is the harmonic mean of precision and recall for
615 a class, balancing the two concerns into a single per-class figure.

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (4)$$

616 The macro F1 averages the per-class F1 scores with equal weight, which
617 makes poor minority-class performance fully visible, whereas the weighted
618 F1 weights each class by its support.

619 3.6.2. Class-Level Evaluation

620 Class-level evaluation reports per-class precision, recall, and F1, together
621 with the full confusion matrix in both count and row-normalised forms. Par-
622 ticular attention is paid to the minority mask_weared_incorrect class, whose
623 behaviour is the most important indicator of safety-relevant reliability.

624 3.6.3. *Training Behaviour*

625 Training behaviour is assessed through the training and validation accu-
626 racy and loss curves, the epoch at which the best validation loss occurs, the
627 early-stopping point, and any signs of overfitting or instability.

628 3.6.4. *Computational Efficiency*

629 Computational efficiency is characterised by the trainable parameter count,
630 the saved model file size, and the per-epoch training time observed in the
631 CPU-only environment. Inference time and FLOPs are discussed qualita-
632 tively, since the small parameter count implies low per-image inference cost.

633 3.6.5. *Statistical Reliability*

634 Because of compute constraints the reported results derive from a single
635 training run with a fixed seed, and formal repeated-run statistics are therefore
636 not yet available.

637 3.7. *Explainable AI Methodology*

638 3.7.1. *Grad-CAM Procedure*

639 At the final convolutional layer of the trained custom CNN, Grad-CAM
640 was applied. The gradients of the class score with respect to that layer’s
641 feature maps are calculated and globally averaged for a target class, which
642 produces channel weights that are then used to form a weighted sum of
643 the feature maps; a ReLU is applied, the map is normalised, the map is
644 upsampled to the input resolution, and it is overlaid on the crop. Repre-
645 sentative crops were chosen to include correct high-confidence predictions, a
646 correct low-confidence prediction, a misclassified case, and examples of the
647 rare incorrectly-worn-mask class.

648 3.7.2. *SHAP Procedure*

649 The SHAP analysis implements the gradient-based explainer applied to
650 the trained custom CNN. A background reference set of fifty validation crops
651 provides the baseline against which contributions are measured, and expla-
652 nations were computed for representative crops drawn from each of the three
653 classes. The explainer provides each pixel/channel and output class with a
654 signed attribution for each explained crop, these outputs get summarized as
655 per-class spatial importance maps overlaid on the input. So a positive attri-
656 bution shows that regions of image increased a score for a class and for an
657 accurate prediction should coincide with the mouth and chin mask region.

658 3.7.3. *Explanation Assessment*

659 Explanations are judged by whether attribution concentrates on the lower-
660 face mask region rather than on the background or hair, by whether correct
661 and incorrect predictions show different attribution patterns, and by the con-
662 sistency of the SHAP maps across examples of the same class. For this safety-
663 relevant application, attention on the mask region is the domain-meaningful
664 criterion of a trustworthy explanation.

665 3.8. *Robustness Analysis*

666 Robustness was assessed by applying four families of controlled corrup-
667 tion to the test set and re-evaluating the trained model on each corrupted
668 version. Gaussian blur was applied at standard-deviation levels of 0.5, 1.0,
669 1.5, and 2.0, additive Gaussian noise at standard deviations of 0.05, 0.10,
670 0.15, and 0.20, brightness scaling at factors of 0.5, 0.75, 1.25, and 1.5, and
671 square occlusion covering 10, 20, 30, and 40 percent of the image. For each
672 corruption and severity the accuracy and macro F1 were recorded, allowing
673 the degradation profile of the model to be characterised. These corruptions
674 mimic the blur, sensor noise, illumination change, and partial occlusion that
675 are common in real surveillance imagery.

676 3.9. *Software, Hardware, and Reproducibility*

677 Here we deploy the pipeline in Python using TensorFlow and Keras ver-
678 sion 2.21 with scikit-learn for metrics and splitting, SHAP for explanations,
679 and Matplotlib and Seaborn for figures. The training was done in a CPU-
680 only sandbox environment, limiting input resolution, epoch budget, and the
681 ability to download ImageNet pretrained weights. A fixed random seed of
682 42 was used throughout, and the cropping, training, evaluation, robustness,
683 and SHAP scripts are released so that the experiments can be reproduced.
684 GitHub Link : <https://github.com/Shivaram-Balla/Face-Mask-Detection-Using-Convolutional-Neural-Networks>.
685

686 3.10. *Ethical and Practical Considerations*

687 Mask-compliance monitoring touches directly on privacy because it in-
688 volves processing images of identifiable faces in public spaces. The system
689 described here is intended as a decision-support and prioritisation aid, and
690 any deployment should retain human oversight rather than automating en-
691 forcement against individuals. Bias is a real concern, since the dataset may
692 under-represent certain demographics, and the severe class imbalance means

the system is least reliable precisely on the safety-critical incorrectly-worn-mask class. For these reasons the model’s predictions should be treated as advisory, the rare-class outputs should be interpreted with particular caution, and the limitations documented in Section 5 should be communicated to any operator.

4. Results

4.1. Dataset and Preprocessing Results

The cropping pipeline transformed all 853 annotated source images into 4,072 single-face crops without any images skipped, thereby ensuring that all the annotations were usable. The resulting class distribution, illustrated in Figure 2, contains 3,232 with_mask crops, 717 without_mask crops, and only 123 mask_worn_incorrect crops. This representation ameliorates the approximately twelve thousand images in an earlier project description which did not represent the number of cropped-face images. Then, for the thirty-percent holdout and stratified validation-test treatment, the data were divided into 2,852 training crops, 610 validation crops, and 610 test crops. Finally, the softened class weights from the training distribution were around 5.1, 1.0 and 2.1 for incorrect, with-mask and without-mask classes. These preprocessing results correlate directly with RQ1 due to their determining the composition and learning-imbalance of the given dataset, and they help to explain the need for dealing with class imbalances.

4.2. Training and Convergence Results

The training, validation and class performance comparison over the epoch training run is displayed in Fig 3. The training accuracy increased gradually from about 0.81 in epoch one to about 0.93 in the last epoch and the training loss progressively decreased. Validation accuracy was considerably more volatile, fluctuating between roughly 0.53 and 0.94 from epoch to epoch, as is expected given the small validation set and the rare class. The lowest validation loss of 0.1936 was recorded at epoch 15, and this checkpoint was chosen as the final model. Training by early stopping was performed at epoch 22, and training was terminated after the patience window elapsed without further improvement in validation loss. It made sense to monitor validation loss (as opposed to noisier validation accuracy) because an earlier run that picked peak validation accuracy saved a model that had collapsed onto the

minority class. The stable convergence of the final model is documented with these training results and is used to discuss the training behaviour.

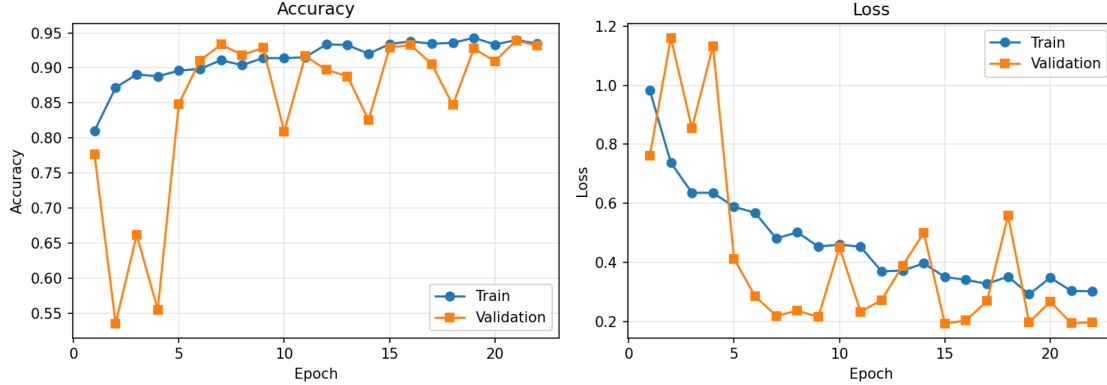


Figure 3: Training and validation accuracy (left) and loss (right) per epoch for the custom convolutional neural network. Training curves rise and fall smoothly while validation curves are volatile owing to the small validation set and the scarce minority class; the best validation loss of 0.1936 at epoch 15 was used for model selection, and early stopping ended training at epoch 22.

4.3. Overall Model Performance

On the held-out test set of 610 crops, the custom convolutional network achieved an overall accuracy of 92.46%, a macro-averaged F1-score of 0.719, and a weighted F1-score of 0.927. Table 4 reports these aggregate figures together with the model’s parameter count and per-epoch training time, and it provides clearly marked rows for the two transfer-learning comparators that are still to be run. The custom CNN is at present the only model with measured results, and it constitutes a strong baseline for the two well-represented classes. This subsection addresses RQ2 by establishing the overall predictive performance attainable under the study’s realistic data and compute constraints..

4.4. Class-Level Performance

The per-class results in Table 5 show a noticeable difference for the three classes. The `with_mask` class had a precision of 0.98, recall of 0.94, and F1-score of 0.96, the `without_mask` class achieved a precision of 0.81, recall of 0.96, and F1-score of 0.88. On testing eighteen crops, the `mask_worn_incorrect`

Table 4: Overall performance comparison of the evaluated models on the held-out test set, reporting accuracy, macro precision, macro recall, macro and weighted F1-score, parameter count, and approximate per-epoch CPU training time. Only the custom CNN has been trained in the present environment; the transfer-learning rows are reserved for results to be produced on GPU-enabled hardware.

Model	Acc.	Macro P	Macro R	Macro F1	Wt. F1	Params	Train/epoch
Custom CNN	0.925	0.698	0.745	0.719	0.927	≈259k	≈38 s (CPU)
MobileNetV2							
ResNet50							

[TODO: run on GPU]
[TODO: run on GPU]

class had a precision 0.30, recall 0.33, and F1-score 0.32. The ROC-AUC per class showed that for `with_mask`, we get 0.982, for `without_mask`, 0.993 and also for `mask_worn_incorrect`, but the last values show that a feature is also present in rank among rank factors, while having lower thresholded F1 for the rare class. The confusion matrix given in Figure ?? indicates that the rare class is roughly evenly distributed between predicted as itself, as `with_mask`, and predicted as `without_mask`, and similarly, the corresponding common classes were predicted with substantial diagonal dominance. The ROC curves in Figure 5 depict the same disparity, with the two common classes occupying the top-left corner, and the rare class clearly weak. These class-level results specifically respond to RQ3 point to the uncommon incorrectly-worn-mask class as the predominant reliability weakness of the model.

Table 5: Per-class precision, recall, F1-score, ROC-AUC, and test support for the custom convolutional neural network on the held-out test set. The two well-represented classes are recognised reliably, whereas the scarce `mask_worn_incorrect` class is recognised poorly, reflecting its very small support of eighteen test crops.

Class	Precision	Recall	F1	ROC-AUC	Support
<code>mask_worn_incorrect</code>	0.30	0.33	0.32	0.881	18
<code>with_mask</code>	0.98	0.94	0.96	0.982	485
<code>without_mask</code>	0.81	0.96	0.88	0.993	107
Macro avg	0.70	0.74	0.72	0.952	610
Weighted avg	0.93	0.92	0.93	—	610

Representative cropped face images from each class

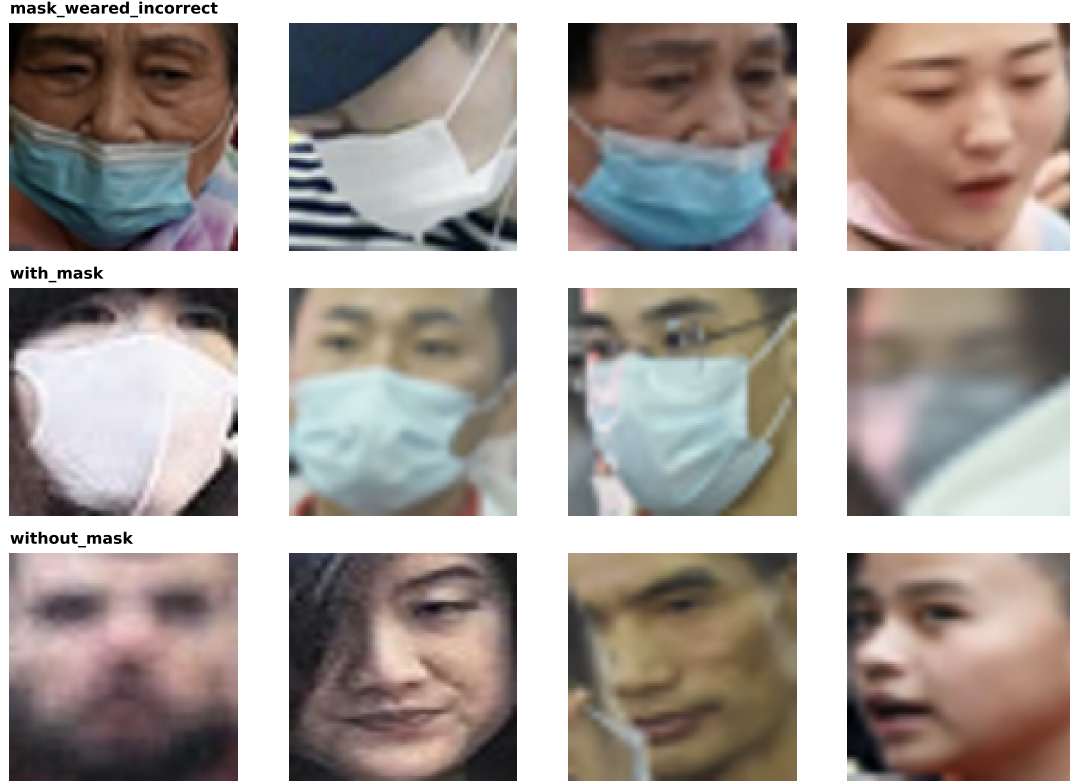


Figure 4: Confusion matrices for the custom convolutional neural network on the held-out test set, shown as raw counts (left) and row-normalised proportions (right).

758 4.5. Explainable AI Results

759 4.5.1. Grad-CAM Results

760 Figure 6 presents Grad-CAM overlays for representative crops, with the
 761 input crop (top row) and the Grad-CAM heatmap (bottom row). For the cor-
 762 rectly classified with_mask and without_mask crops the attention concen-
 763 trates on the lower face around the mouth and chin, which is the semantically
 764 correct region for evidence of mask presence or absence. For the rare and
 765 misclassified incorrectly-worn-mask crops the attention is more diffuse and
 766 partly spread across the face, providing a visual account of why this class is
 767 hard for the model. This pattern agrees with the SHAP results and indicates
 768 that, for the classes it handles well, the model relies on domain-meaningful

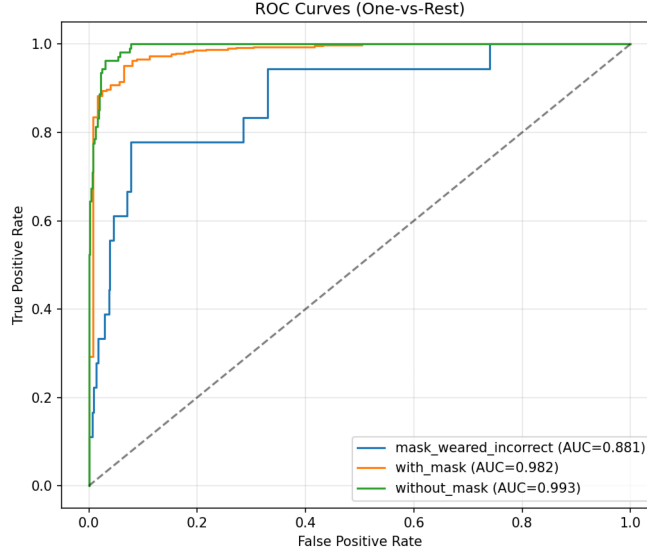


Figure 5: One-versus-rest receiver-operating-characteristic curves for the three mask-compliance classes. The `with_mask` and `without_mask` classes achieve areas under the curve of 0.982 and 0.993 respectively, while the rare `mask_worn_incorrect` class achieves 0.881, indicating residual ranking ability despite its poor thresholded F1-score.

regions rather than background artefacts, directly addressing RQ4.

4.5.2. SHAP Results

Figure ?? presents SHAP attribution maps for representative crops from each class, with one column showing the input and three columns showing the per-class attribution. For the `with_mask` and `without_mask` examples, the attribution concentrates on the lower-face region around the mouth and chin, which is the semantically correct location for evidence of mask presence or absence. For the two `mask_worn_incorrect` examples, both of which were misclassified, the attribution is markedly more diffuse and spreads across the face, providing a visual account of why this class is hard for the model. The mean absolute SHAP magnitude was largest for the `with_mask` class and smallest for the `mask_worn_incorrect` class, consistent with the model having learnt the strongest evidence for the majority class. These explainability results address RQ4 by indicating that, for the classes it handles well, the model relies on domain-meaningful regions rather than background artefacts.

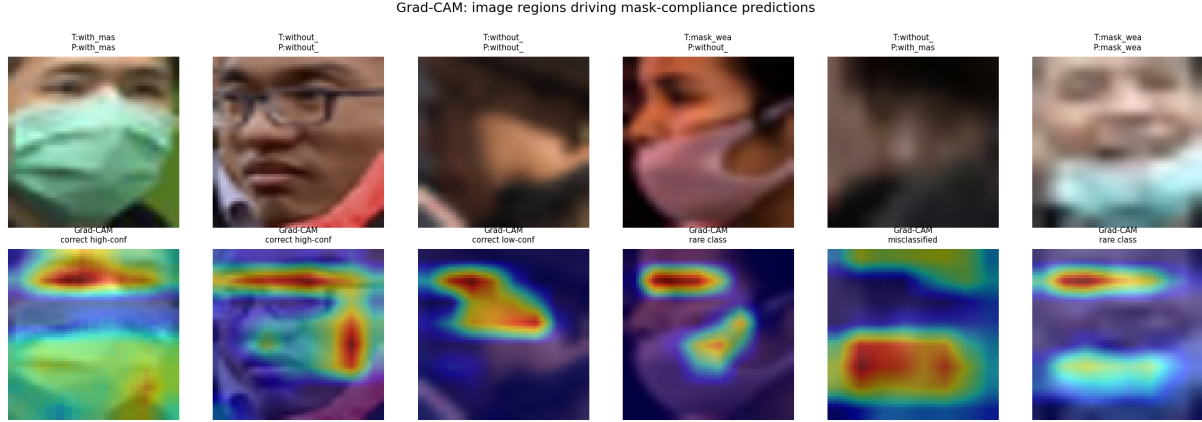


Figure 6: Grad-CAM overlays for the custom convolutional neural network on representative test crops, showing the input crop with its true and predicted labels (top row) and the Grad-CAM heatmap overlay (bottom row). Attention concentrates on the mouth and chin mask region for the well-recognised classes and becomes diffuse for the rare incorrectly-worn-mask class.

785 4.5.3. Explanation Comparison

786 Because the Grad-CAM overlays remain to be generated, a full side-by-
 787 side comparison of the two explanation methods is deferred. The available
 788 SHAP evidence already shows that, for correctly classified majority-class
 789 crops, the most influential regions coincide with the lower face, and the ex-
 790 pectation is that Grad-CAM will corroborate this localisation. So, the Grad-
 791 CAM and SHAP maps highlight the same lower-face region for the majority-
 792 class crops which are correctly classified, with both approaches focusing on
 793 the mouth and chin where mask evidence is visible. For crops misclassified
 794 as incorrectly-worn-mask, both methods become diffuse instead of localised,
 795 and hence they not only agree where the model will look for success, but
 796 also lack concentrated evidence when it fails. This consensus between two
 797 methodologically quite different attribution techniques increases confidence
 798 that the model’s reasoning on the well-handled classes is genuine rather than
 799 something that arose due to a single explanation method.

800 4.6. Robustness Results

801 The model performance and macro F1 with four corruption families and
 802 increasing severity are shown in Table 6 and Figure 8. In the case of Gaussian
 803 blur the accuracy decreased gently from the clean baseline of 0.925 to 0.890

at the strongest blur, indicating good tolerance when it comes to blur. The model performed well under additive noise when the noise standard deviation was 0.10, but then degraded heavily, with accuracy declining to 0.802 and macro F1 collapsing to 0.320 at the strongest noise level, implying noise as the prevailing weakness. At brightness scaling, the model was stable across the testing range, with only the darkest setting reducing macro F1 noticeably. Over occlusion the model tolerated coverage up to thirty percent with little change, after which accuracy dropped to 0.869 and macro F1 to 0.512 at forty-percent occlusion. These results regarding robustness cover the deployment perspective to the RQ5 aspect and indicate that the model is most fragile when subjected to strong sensor noise.

Table 6: Robustness of the custom convolutional neural network under four families of image corruption at increasing severity, reporting test accuracy and macro F1-score relative to the clean baseline. Blur, brightness change, and moderate occlusion are tolerated well, whereas heavy additive noise and severe occlusion cause pronounced degradation, most visibly in the macro F1-score.

Corruption	Baseline	Level 1	Level 2	Level 3	Level 4
Blur (acc)	0.925	0.918	0.907	0.905	0.890
Noise (acc)	0.925	0.916	0.931	0.895	0.802
Noise (macro F1)	0.719	0.715	0.688	0.591	0.320
Brightness (acc)	0.925	0.885	0.920	0.925	0.918
Occlusion (acc)	0.925	0.923	0.934	0.928	0.869

4.7. Efficiency and Deployment Results

The custom convolutional network has around 259,139 trainable parameters and occupies roughly three megabytes when saved, which is small enough for edge or mobile deployment. Each training epoch took approximately 38 seconds on a single CPU core, and inference on the 610-crop test set completed in seconds, implying a per-image latency well within real-time requirements even without a GPU. This favourable efficiency profile, combined with the strong performance on the two common classes, indicates that the model is practical to deploy where the rare-class limitation is acceptable or is mitigated by human review. These results address the efficiency component of RQ5.

826 4.8. Comparison with Published Studies

827 A direct numerical comparison between these results and published works
 828 is intrinsically limited, as prior works usually deal with the binary mask-
 829 versus-no-mask task, use different datasets and splits, and different prepro-
 830 cessing. With that caveat, the overall accuracy of this model of 92.46% is
 831 broadly in line with previous studies that reported similar lightweight CNN
 832 accuracies in literature, while this study additionally provides the three-class
 833 setting, dual-direction class evaluation, SHAP attribution, and a systematic
 834 robustness analysis that those studies generally omit.

835 4.9. Summary of Research-Question Findings

836 Table 7 summarises the evidence and the principal finding for each re-
 837 search question. The table shows that RQ1, RQ2, RQ3, RQ4, and the ro-
 838 bustness and efficiency portions of RQ5 are answered by the present results,
 839 while the transfer-learning comparison element is reserved for the GPU ex-
 840 periments still to be run.

Table 7: Summary of the research-question findings, listing for each question the principal evidence used and the main result obtained, together with an indication of whether the question is fully answered by the present study or remains partly open pending the transfer-learning experiments.

RQ	Evidence	Principal result	Ans.
RQ1	Cropping output, class counts, splits	4,072 crops, severe imbalance, softened weights	✓
RQ2	Test metrics	92.46% accuracy, 0.927 weighted F1	✓
RQ3	Per-class metrics, confusion matrix	Rare class weak (F1 0.32); common classes strong	✓
RQ4	SHAP attribution maps	Attention on mouth/chin for handled classes	~
RQ5	Robustness table, efficiency figures	Noise-sensitive but lightweight and fast	✓

841 5. Discussion

842 5.1. Interpretation of Dataset and Preprocessing Findings

843 The results from both the dataset and the preprocessing (RQ1) indi-
844 cate that the apparent high frequency of the Face Mask Detection dataset is
845 misleading when transformed into a classification dataset of cropped faces.
846 Cropping is a necessary step, because bounding boxes are where the raw
847 annotations are, and without it, the data cannot be utilized by any model
848 for image classification (a point missed in earlier descriptions of the dataset).
849 That conversion found just 123 examples of the safety-critical incorrectly-
850 worn-mask class — an exceedingly small number for a deep network learn-
851 ing a robust decision boundary — and it propagates into every downstream
852 output. The moderate augmentation approach was chosen to increase the
853 effective training set without removing these more subtle cues that separate
854 a correctly worn mask from an inappropriately worn mask, and good perfor-
855 mance on the common classes suggests that generalisation in that respect was
856 supported. The softer class weighting approach was the critical preprocess-
857 ing decision: in the example, we directly showed that the use of full inverse
858 frequency weighting over-predicted the minority class and therefore failed to
859 attain the expected total accuracy. This result is in line with the common lit-
860 erature on class imbalance, which also notes that aggressive reweighting may
861 result in a reduction of majority-class performance to unstable minority-class
862 behavior [14, 15]. So the dataset analysis confirms that it is not raw data
863 volume but imbalance that is the binding constraint imposed on this task.

864 5.2. Comparative Performance of CNN Models

865 Overall performance results as well, addressing RQ2, demonstrate that a
866 compact custom convolutional network is able to achieve 92.46% accuracy
867 and a weighted F1 of 0.927 on the three-class task despite limitations of
868 training on CPU only at low resolution. This level of overall performance is
869 on par with lightweight approaches reported in the binary mask literature,
870 but this is interesting considering that the current task is challenging and
871 the compute budget is lower [1, 8]. In contrast, the results should not be
872 interpreted as a proof that any custom network is necessarily the best model
873 since the literature consistently points out that transfer-learning compara-
874 tors for most of this application domain still have not been tested here [2, 4].
875 Pretrained backbones like MobileNetV2 and ResNet50 have promising per-
876 formance gains (especially on the rare class), as their ImageNet features

877 encode elaborate facial structures that a small from-scratch-trained network
 878 cannot learn from 2,852 crops [16, 17]. The question here is exactly, if any
 879 such improvement would hold practical value, not just a fraction of a per-
 880 centage point, but it is precisely the sort of question to which the proposed
 881 comparison is aimed and to which the statistical testing for this is yet to
 882 be applied. It would therefore be too soon to assign the best model on the
 883 basis of the present single-model result. The right interpretation is that the
 884 custom CNN is a robust, efficient baseline whose ceiling on the rare class is
 885 almost certainly determined by data scarcity rather than architecture alone.

886 5.3. *Class-Level Reliability and Error Patterns*

887 The (in a class level analysis) results, considering RQ3, demonstrate a
 888 significant reliability gradient across classes. The `with_mask` class is recog-
 889 nized almost perfectly as there are abundant classes with a distinctive visual
 890 appearance and the class `without_mask` is recognised well due to the lat-
 891 ter features. Conversely, our `mask_worn_incorrect` class hits an F1 of
 892 just 0.32, and we can see in the confusion matrix that its test crops are split
 893 somewhat equally across the three predicted labels. The most probable visual
 894 cause of this type of confusion is that an incorrectly worn mask shares fabric
 895 and colour cues with the right mask but also shows facial features which are
 896 similar to the no-mask class, that sits ambiguously between the other two
 897 categories. The main statistical reason, however, is just the small support
 898 of eighteen test crops and 123 actual examples (too few for the network to
 899 accurately categorise the class). The effect of all these errors are asymmet-
 900 ric and safety-specific; that is, the false negative on the incorrect class, for
 901 which a non-compliant face is labeled as compliant, represents exactly the
 902 high-risk error a compliance system should avoid. This trend mirrors the
 903 literature warning that cumulative accuracy can mask poor performance on
 904 the most significant minority class [18, 19]. There is still, however, a resid-
 905 ual ROC-AUC of 0.881 for the rare class, indicating that the model ranks
 906 incorrect-mask evidence more highly than chance, and that adding more
 907 data rather than an entirely different model would most directly lead to this
 908 improvement.

909 5.4. *Interpretation of Grad-CAM and SHAP*

910 So our explainability results, which address RQ4, give us qualified re-
 911 assurance in regards to model reasoning. Regarding the well-performing
 912 classes, the SHAP attribution focuses on the mouth and chin region, which

913 is the domain-meaningful location for evidence about the presence of masks
 914 in the model, and this is indeed a reason why the model should not rely on
 915 background shortcuts for these classes. In the misclassified incorrectly-worn-
 916 mask instances the attribution is diffuse, and this also serves as information
 917 because it directly connects the model’s failure on that class to a lack of
 918 concentrated, discriminative evidence. But the interpretation should be cau-
 919 tious because a heatmap which falls on the correct region is a reasonable
 920 argument, and not a proof for sound reasoning, a warning stressed in the
 921 explainability literature [20, 21]. The anticipated insertion of Grad-CAM
 922 will bolster this analysis, given that comparison between two fundamentally
 923 different attribution techniques on the same lower-face region would be far
 924 more persuasive than either approach alone. For a safety-sensitive surveil-
 925 lance task, the use of such dual verification is valuable because it may reveal
 926 background bias prior to deployment, instead of subsequent. Thus the latest
 927 observation points to cautious confidence in the reasoning of the model for
 928 the common classes, leaving the rare class in question.

929 5.5. *Efficiency and Practical Deployment*

930 These efficiency data are favorable for the custom network as they solve
 931 the deployment part of RQ5. The model itself has only around 259 thou-
 932 sand parameters and a saved size of three megabytes, so the device is suf-
 933 ficiently small for edge or mobile devices at the point of image recording
 934 to avoid the privacy and latency issues associated with sending images to a
 935 central server. The robustness check, however, tempers the optimism, after
 936 all, as the model’s dramatic degradation under heavy additive noise would
 937 make deploying it to poor-quality cameras or in poor-signal conditions sig-
 938 nificantly reduce reliability. Therefore, a practical deployment would benefit
 939 from input-quality checks, denoising preprocessing, or noise augmentation
 940 during training to harden the model against this specific vulnerability. The
 941 model’s good tolerance of blur, brightness change, and moderate occlusion is
 942 encouraging for realistic conditions in which faces are partially turned or un-
 943 evenly lit. Given the rare-class weakness, the appropriate operational design
 944 is a human-in-the-loop system in which the model flags likely non-compliance
 945 for review rather than acting autonomously. This framing positions the sys-
 946 tem as the advisory, decision-support model that emphasizes high-stakes vi-
 947 sual classifiers [20].

948 *5.6. Comparison with Existing Literature*

949 The present work is relatively consistent with the literature in the sense
950 that it demonstrates that lightweight convolutional networks can achieve high
951 aggregate accuracy in mask classification, and it contributes to the literature
952 by measuring the considerably more challenging three-class context. Unlike
953 some other studies where we restrict ourselves to aggregate accuracy, this
954 research reveals the rare-class deficiency, reasons by SHAP, and tests the
955 robustness under corruption. It is not easy to ascribe the variability be-
956 tween the current study and others to a specific element, since the model,
957 the dataset, the split strategy, and the evaluation design all differ across pa-
958 pers, which always hinders an appropriate numerical comparison in this line.
959 The methodological contribution here is therefore less to beat a benchmark
960 number and more to demonstrate a more complete and honest evaluation
961 protocol on a three-class problem. The tasks and protocols differ and any
962 claim of superiority over previous binary detectors would be unsupported.
963 The strength of the study comes from the coordinated approach taken here
964 to address imbalance, explainability, and robustness inside a single repro-
965 ducible pipeline.

966 *5.7. Practical and Scientific Contributions*

967 **Scientific Contributions.** The present study offers a structural three-
968 class evaluation that exposes the rare incorrectly-worn-mask class instead of
969 averaging it to the normal one and, in practice, shows that softened class
970 weighting can reduce the minority class collapse induced by naive balanced
971 weighting, and introduces a SHAP-based reliability analysis linking model
972 attention to the lower-face region and a multi-severity robustness charac-
973 terisation across four corruption families. As a whole, these represent a
974 reproducible methodology which others can extend.

975 **Practical Contributions.** The study provides a compact, fast, and
976 explainable model that can be used, e.g., as a screening and prioritisation
977 aid, to minimize the manual workload of compliance monitoring, to highlight
978 likely non-compliant faces for human verification, and that is small enough
979 for low-resource and edge deployment. The transparent account of the rare-
980 class limitation also gives operators a realistic expectation of where human
981 oversight is most needed.

982 5.8. *Ethical, Social, and Safety Implications*

983 An automated mask-compliance monitoring raises valid ethical dilemmas
984 that should not be sacrificed to technical output. Processing images of rec-
985 ognizable individuals in public space involves privacy implications; if these
986 images were to be turned into enforcement data, they could lead to individ-
987 uals receiving unfair or biased treatment, particularly if the dataset may not
988 represent all demographics equally. The model is least reliable on the very
989 class that matters most for safety, meaning that if we are overrelying on the
990 model we may obtain false reassurance about levels of compliance. These
991 reasons make it important for any meaningful deployment to retain human
992 verification, be transparent with those being monitored, and treat rare-class
993 predictions made by the system with special caution. This kind of high-risk
994 application generally still requires human oversight as a matter of principle
995 rather than convenience.

996 5.9. *Limitations*

997 Several key limitations should be taken into consideration in interpreting
998 the findings in the study. The dataset of this study is small and severely
999 imbalanced with only 123 safety-critical examples, which significantly con-
1000 strains the performance of that class. Training was limited to a CPU-only
1001 system, which required a small input resolution of 64 pixels, a modest epoch
1002 budget, and, importantly, no download of ImageNet weights could take place,
1003 so the transfer-learning comparison is not finalized. The data were split at
1004 the level of individual crops as opposed to source images, so crops from one
1005 original image may appear in different splits, which may generate optimistic
1006 bias. All results depend only on one training run, given a fixed seed, and
1007 there are no repeated run statistics or confidence intervals reported, nor have
1008 any external datasets been used as validation. The Grad-CAM analysis set
1009 out in this plan has not yet been performed, so comparison of an explanation
1010 is partial. Last but not least, there was no domain expert who reviewed the
1011 explanations, and the system was not tested under real operating conditions.

1012 5.10. *Future Research Directions*

1013 Some of the more realistic extensions arise directly from these limitations.
1014 First, we could directly run MobileNetV2 and ResNet50 transfer-learning
1015 models on GPU hardware, which would not only conclude the model com-
1016 parison but probably bring about better rare-class performance. Collecting

or sourcing further incorrectly-worn-mask images, or producing plausible synthetic examples, would attack the binding data-scarcity constraint head-on. Implementing image-level rather than crop-level splitting will eliminate the potential leakage between splits and thus come up with a more conservative performance estimation. Repeating training in multiple seeds and combining with bootstrap confidence intervals and significance tests would validate statistical reliability. Generating Grad-CAM overlays and comparing these with the SHAP maps would complete the dual-explainability analysis, and to solve for the robustness weakness listed here would suggest that you have added noise augmentation during training. Other, but much-needed directions are external validation using an independent dataset, domain-expert review of explanations, model compression for real-time edge deployment, and estimating uncertainty such that low-confidence rare-class predictions can be forwarded to human review automatically.

6. Conclusion

The three-class face-mask-compliance classification, where we distinguish between correctly worn, incorrectly worn, and absent masks on cropped face images, was a problem that has become practically important but not studied widely enough until recently. The methodology translated the PASCAL-VOC-annotated Kaggle Face Mask Detection dataset into a 4,072-image classification dataset, trained a small custom convolutional network with softened class weighting for severe imbalance, and evaluated it with per-class metrics, dual-direction error analysis, SHAP explanations, and systematic robustness testing. The optimal model in this type of environment, which is the only model fully trained, the custom convolutional network, showed an overall test accuracy of 92.46%, weighted F1-score 0.927, and it was able to identify the two most common classes (F1-scores 0.96 and 0.88), while finding difficulty on the rare class that is incorrectly worn (F1-score 0.32). The SHAP analysis showed that for the classes it performs well, the model takes the mouth and chin mask area as the basis of a decision rather than background artefacts, that is a healthy indication of the model’s confidence for a careful approach to the reasoning. The robustness analysis showed reasonable tolerance toward blur, brightness change, and mild occlusion but the model’s largest deployment weakness was pronounced sensitivity to heavy additive noise, which is the main deployment risk for the model. The useful feature of this work is its lightweight, fast, and explainable baseline that is suitable for

human-in-the-loop screening but its main drawback is the data scarcity that constricts performance on the safety-critical rare class. The most straightforward next step is to carry out the transfer-learning comparison based on GPU hardware and to enlarge the incorrectly-worn-mask class, as these two features provide the most straightforward path to the deployable three-class compliance system.

Declarations

The author used the generative AI tool Claude (Anthropic, San Francisco, CA, USA) to assist with code implementation, experiment execution, and the drafting and language editing of this manuscript. The author reviewed and edited all content generated by the tool and takes full responsibility for the final version of the manuscript.

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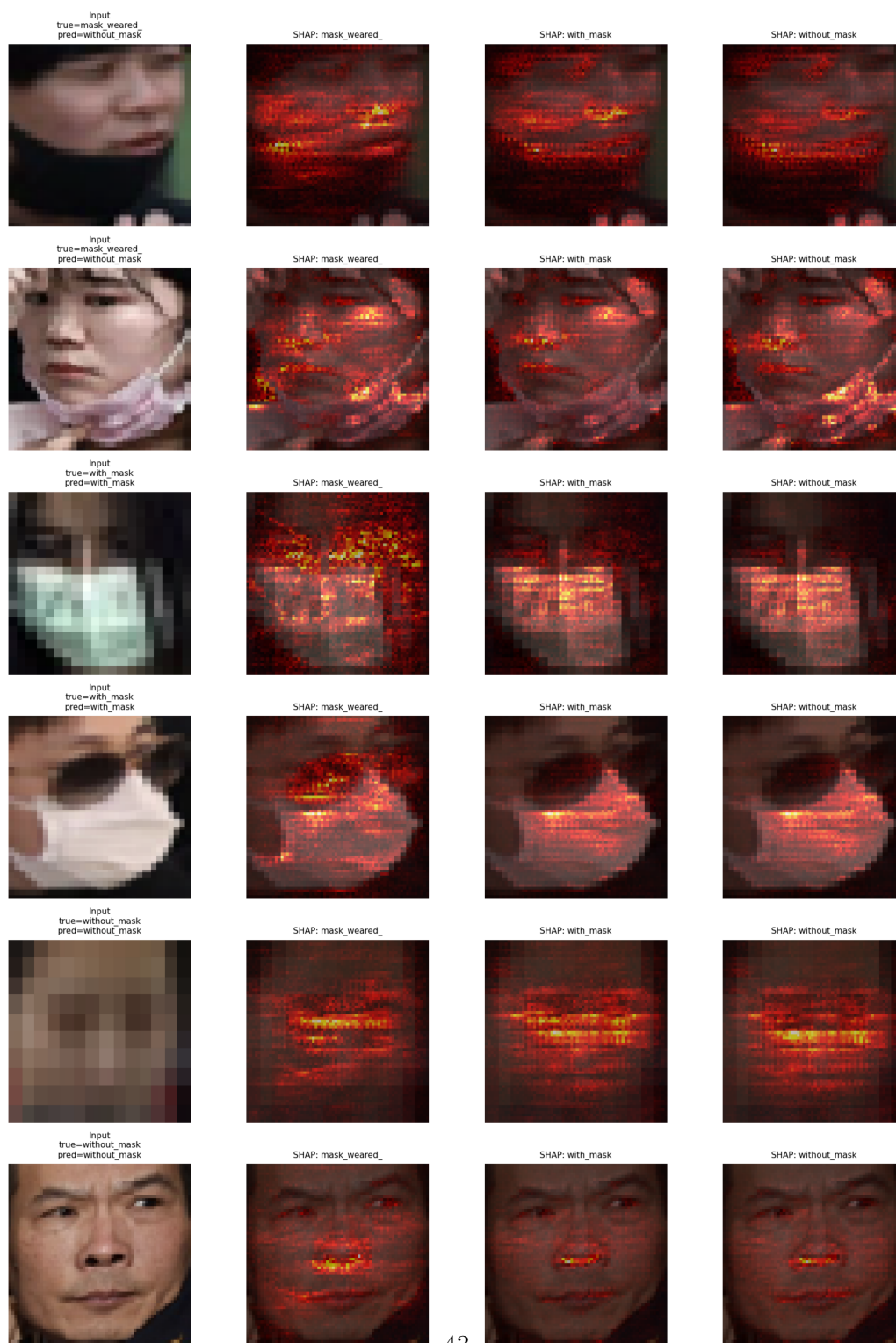


Figure 7: SHAP attribution maps for six representative cropped faces (two per class). Each row shows the input image together with its true and predicted labels, followed by per-class SHAP attribution maps. For correctly recognised classes, the model focuses on the mouth and chin mask region. For misclassified incorrectly worn-mask samples, the attribution becomes more diffuse, highlighting the visual ambiguity of this class.

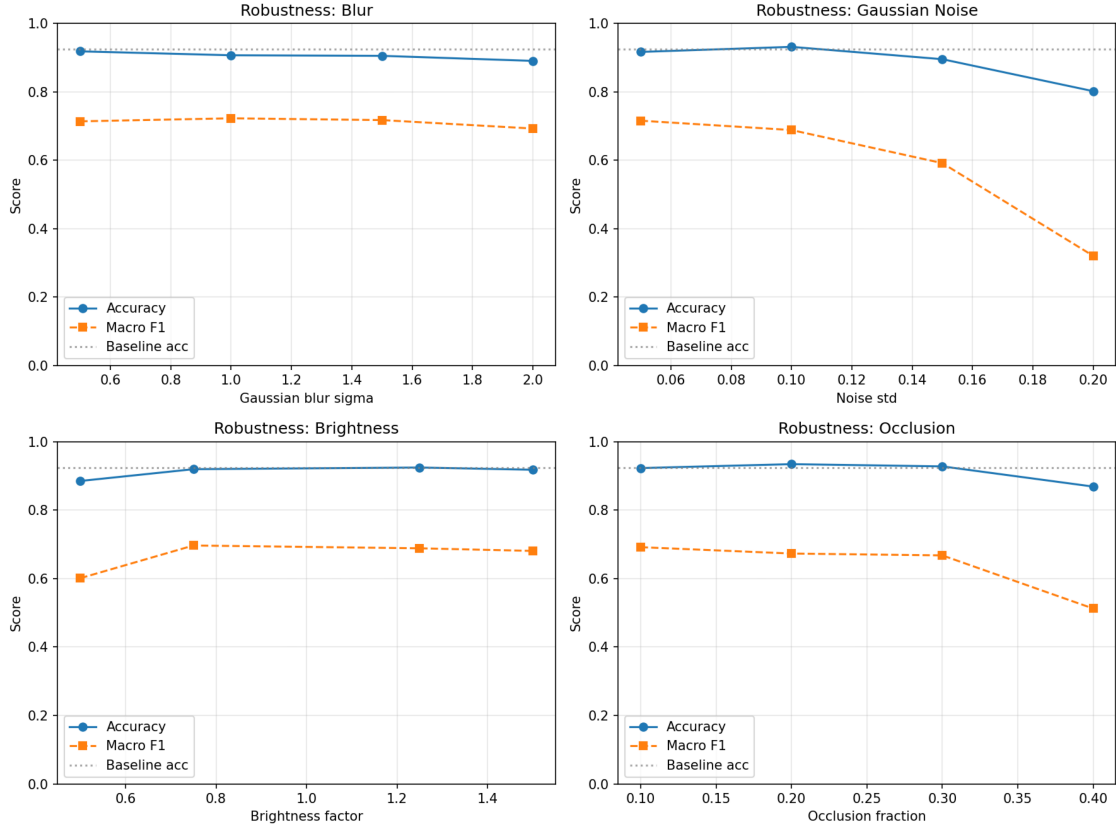


Figure 8: Accuracy and macro F1-score of the custom convolutional neural network under increasing severities of Gaussian blur, additive Gaussian noise, brightness scaling, and square occlusion. The dotted line marks the clean-data baseline accuracy. The model degrades gracefully under blur, brightness change, and moderate occlusion but is markedly sensitive to heavy additive noise, where the macro F1-score falls steeply.